Performance Testing

Date	26 September 2026
Team ID	SWTID-2026-3976
Project Name	AI FAQ Assistant API (AnswerBot)
Maximum Marks	5 Marks

Performance Testing

Response times of the key endpoints were recorded with Postman on the local development server against the MongoDB Atlas database. The figures below are the values observed in those Postman runs (single user, one request per endpoint); no separate load-testing tool was used.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman
Type of Testing	Functional response-time check (single virtual user)
Target Module / API	Authentication (register, login, profile) and AI answer endpoint
Test Environment	Local server (http://localhost:5001) with MongoDB Atlas
Test Date	26 September 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	POST /api/auth/register - create a new user	1	< 1	201 Created with JWT
2	POST /api/auth/login - log in with valid credentials	1	< 1	200 OK with JWT
3	GET /api/auth/profile - fetch profile with Bearer token	1	< 1	200 OK with profile data
4	POST /api/ai/answer - ask "what is mongodb"	1	~3	200 OK with AI answer

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	847 ms (average of the 4 requests: 664, 115, 48 and 2,560 ms)	Pass	Auth requests are fast; AI adds Gemini latency
2	Response Time (Max)	< 5 seconds	2.56 s (AI answer)	Pass	Slowest call is the AI endpoint
3	Throughput (Req/sec)	-	Not measured	-	Load test not run in this phase
4	Error Rate	< 1%	0% (4 of 4 requests successful)	Pass	All responses were 200 or 201
5	CPU Utilization	< 80%	Not measured	-	Planned with a load-testing tool

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
6	Memory Utilization	< 80%	Not measured	-	Planned with a load-testing tool

Step 4: Observations & Analysis

- Login (115 ms) and profile (48 ms) are very quick because they use indexed lookups and stateless JWT verification.
- Registration takes longer (664 ms) because bcrypt hashing is deliberately slow.
- The AI answer (2.56 s) is dominated by the Google Gemini call and is still inside the 5 s target.
- Next step: run JMeter, k6 or Locust load tests to measure throughput, CPU and memory under concurrent users.

Step 5: Screenshots / Evidence

Postman results for the four tested requests:




